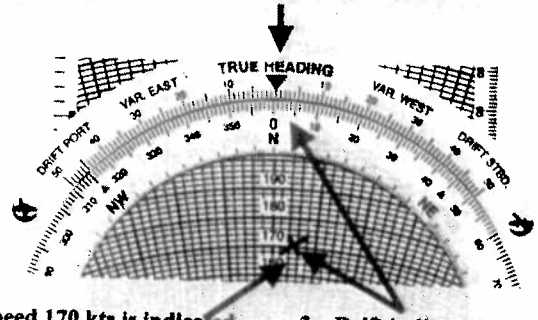
1. Required heading to steer on the window disc (000.5°)is indicated by the true heading index mark



2. Ground speed 170 kts is indicated by the cross

3. Drift indicated by cross (2.5 right) is the same s that indicated by the required track (003) on the window aligned with (2.5 right) on then outer scale.

<u>CRP5 71 c.</u>

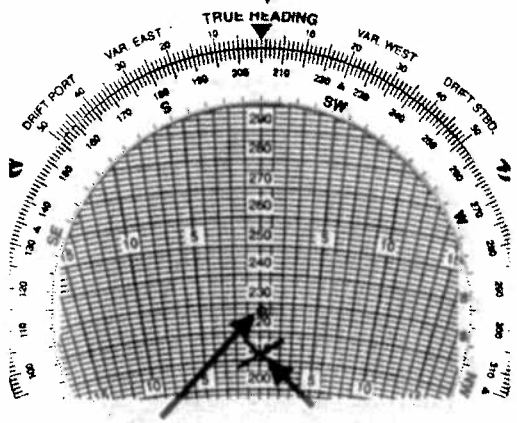
To solve this type of problem on the CRP5:

a. Rotate the window so that the wind direction (205) aligns with the true heading index.

b. Using the low speed side of the slide, draw a cross on the window directly below the centre dot to represent the wind speed (15 kts).

c. Move the slide up or down until the TAS (227 kts) aligns with the centre dot. This situation is illustrated below.

1. Rotate window to align wind direction (205) with true heading index mark



2. Set centre dot to align with 227 TAS

3. Draw a cross below the centre dot to represent 15 kts wind speed

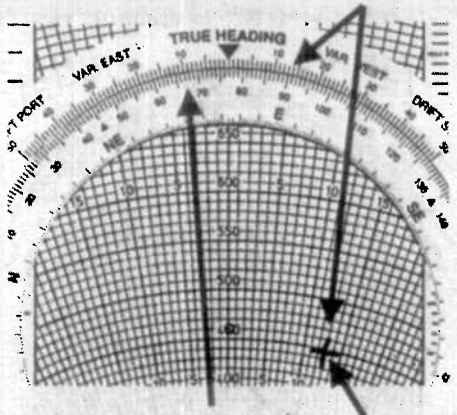
d. Rotate window to align track (316) with true heading index mark.

e. Read off the drift at the cross. In this case it is 3.5° right drift.

f. Rotate the window to align the track (316) with the drift 3.5 right on the outer scale. Recheck the drift indicated by the cross. In this case it is still approximately 3.5° right. The drift indicated by the cross and that indicated where track 316 aligns with the outer scale are both is 3.5 right. The drift is now said to be balanced.

g. Read off the heading to steer on the window disc opposite the true heading index mark and the ground indicated by the cross. In this case it is 312.5° 231 kts. This is closest to 312° 232 kts (option c). This situation is illustrated below.

1. Drift balanced with both 090° and cross aligned with about 13° drift.



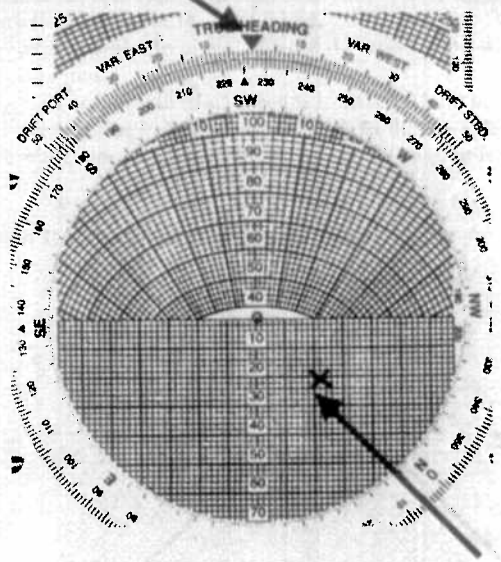
2. 10° E variation aligned with 067°.
3. Adding 20 deviation gives 0690.
4. Cross indicates 449 kts ground speed.

CRP5 87 b.

To solve this type of problem using the CRP5:

a. Wind directions provided by ATIS and runway directions are both given in terms of magnetic headings. The variation can therefore be ignored in this question.
b. Set the low speed slide so that the centre dot is at the top of the square grid at the base of the slide.
c. Set wind direction 180° against the true heading index mark.
d. Draw a cross 30 kts below the centre dot to represent the wind.
e. Align the runway heading 227° with the true heading index mark.
f. Read off 22 kts (option a) crosswind component by measuring the horizontal distance from the centre line of the grid to the cross. These settings are illustrated below.

1. Runway heading 227° set against true heading index mark.



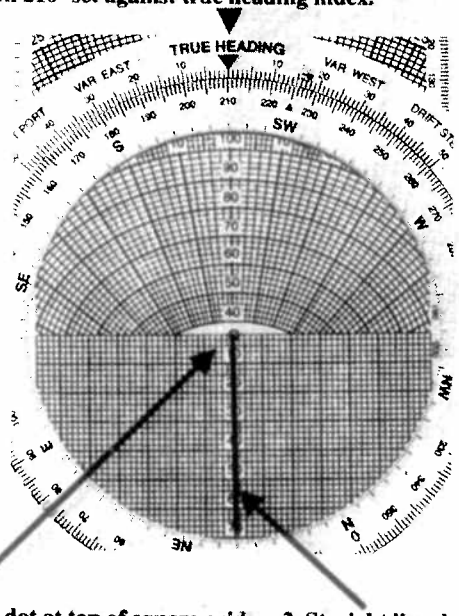
2. Horizontal distance from centre line to cross indicates 22 kt crosswind.

CRP5 88 d.

To solve this type of problem using the CRP5:

a. Both the wind direction given by ATIS and the runway direction are specified as magnetic bearings, so the variation can be ignored.
b. Using the low speed side of the slide set the centre dot at the top edge of the square gridded area at the base of the slide.
c. Rotate the window to align the wind direction 210° with the true heading index mark.
d. Draw a vertical line down the centre from the centre dot to the bottom of the window. This situation is illustrated below.

1. Wind direction 210^0 set against true heading index.



2. Centre dot at top of square grid. 3. Straight line drawn below dot

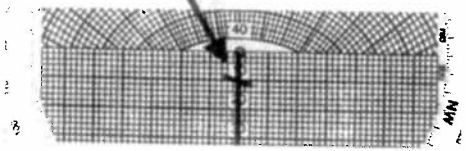
e. Rotate window to align runway direction 047^0 with true heading index mark.

f. Moving the slide upwards, align the centre dot with the 40 kts arc. Draw a cross at the point where the previously drawn straight line. Move the slide back down to its original position with the centre dot at the top of the square gridded area.

g. Rotate the window to position the cross on the centre line directly below the centre dot.

h. Measure the distance down from the centre dot to the cross. This is approximately 11 kts (option d). this situation is illustrated below.

1. Cross is approximately 11 kts below centre dot.



CRP5 89 a.

To solve this type of problem using the CRP5:

a. Set the low speed side of the slide so that the top line of the squared section aligns with the centre dot in the window.

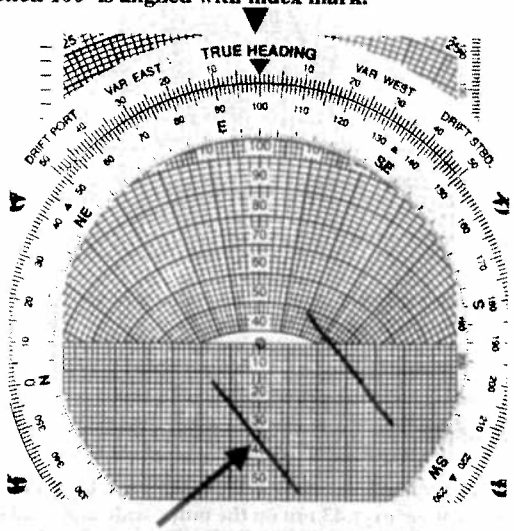
b. Rotate the window to align the runway direction (063).

c. Draw vertical lines to represent the maximum crosswind component 20 kts to the left and right of the centre line.

d. Rotate the window disc to align the wind direction (100) with the true heading index mark.

e. The maximum acceptable wind speed is indicted where one of these lines crosses the centre line below the centre dot. In this case the maximum allowable wind speed is 33 kts (option a). This situation is illustrated below.

1. Wind direction 100^0 is aligned with index mark.



2. Maximum acceptable wind speed (33 kts) is indicated where one of the lines crosses the centre line below the centre dot.

CRP5 90 d.

To solve this type of problem using the CRP5:

a. Using the high speed slide set 450 kts TAS at the centre dot.

b. Set wind direction 200^0 against the true heading index mark.

c. Draw a cross on the centre line 60 kts below the centre dot.

d. Align course 250^0 with true heading index mark.

e. The cross now indicates 6.5^0 right drift.

f. Align course 250^0 with 6.5^0 right drift and read off 6.0^0 right drift at the cross.

g. Set course 250^0 at 6.0^0 right drift and again read of drift at the cross.

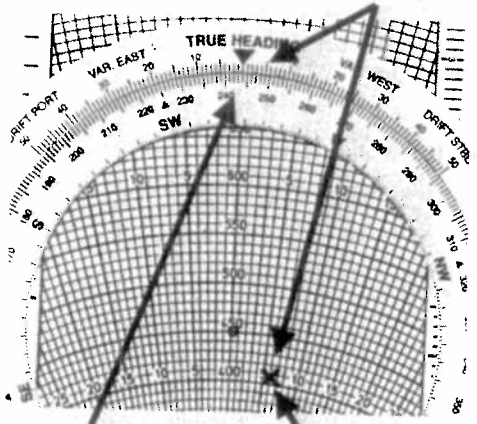
h. Repeating this process will eventually give approximately 6.0^0 drift both on the outer scale and at the cross. The drift is now said to be balanced.

i. Read off the heading 244^0 opposite the true heading index mark.

j. Read off the ground speed 407 kts at the cross.

k. Flight time = distance (315 nm) divided by ground speed (407 kts) = 46.4 min. ETA at B = ETD at A (0650) + 46.4 min = 0736.4, or approximately 0736 (option d). The above settings are illustrated below.

1. Drift balanced with both 250° and cross aligned with about 6° drift.



2. True heading index aligned with 244°.

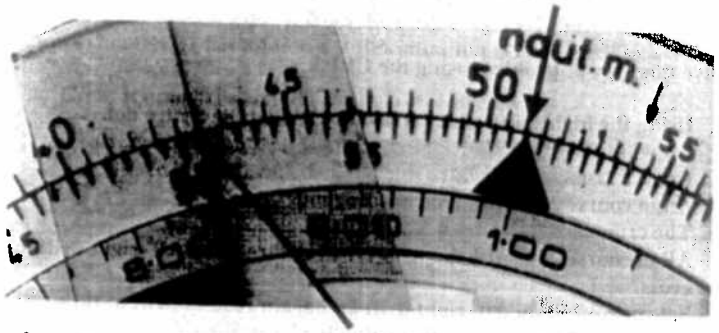
3. Cross indicates 407 kts ground speed.

CRP5 91 a.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 510 kts on the outer scale.
b. Position the cursor over 43 nm on the outer scale and read off 5.5 min where cursor crosses the inner scale. This is closest to 5 min (option a).
c. These settings are illustrated below.

1. 60 min index aligned with 510 kts.



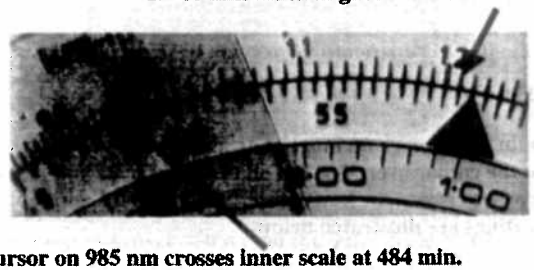
2. Cursor on 43 nm crosses inner scale at 5.5 min.

CRP5 92 b.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 122 kts on the outer scale.
b. Position the cursor over 985 nm on the outer scale and read off 484 min where cursor crosses the inner scale. This 8 hr 04 min (option b).
c. These settings are illustrated below.

1. 60 min index aligned with 122 kts.



2. Cursor on 985 nm crosses inner scale at 484 min.

CRP5 93 d.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 236 kts on the outer scale.
b. Position the cursor over 354 nm on the outer scale and read off 90 min where cursor crosses the inner scale. This 1 hr 30 min (option d).
c. These settings are illustrated below.

1. 60 min index aligned with 236 kts.

2. Cursor on 354 nm crosses inner scale at 90 min.

CRP5 94 d.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 435 kts on the outer scale.
b. Position the cursor over 1920 nm on the outer scale and read off 265 min where cursor crosses the inner scale. This 4 hr 25 min (option d).
c. These settings are illustrated below.

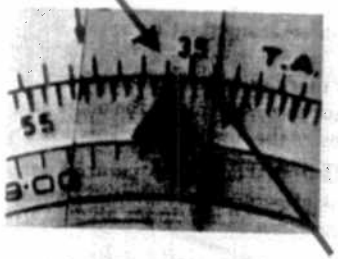
1. 60 min index aligned with 435 kts.

2. Cursor on 1920 nm crosses inner scale at 265 min.

CRP5 95 d.
To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 345 kts on the outer scale.
b. Position the cursor over 3560 nm on the outer scale and read off 619 min where cursor crosses the inner scale. This 10 hr 19 min (option d).
c. These settings are illustrated below.

1. 60 min index aligned with 345 kts.

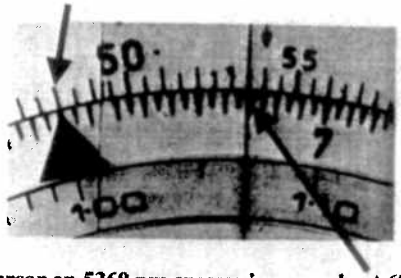


2. Cursor on 3560 nm crosses inner scale at 619 min.

CRP5 96 a.
To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 480 kts on the outer scale.
b. Position the cursor over 5360 nm on the outer scale and read off 670 min where cursor crosses the inner scale. This 11 hr 10 min (option a).
c. These settings are illustrated below.

1. 60 min index aligned with 480 kts.



2. Cursor on 5360 nm crosses inner scale at 670 min.

CRP5 97 c.
To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 95 kts on the outer scale.
b. Position the cursor over 480 nm on the outer scale and read off 303 min where cursor crosses the inner scale. This 5 hr 03 min (option c).
c. These settings are illustrated below.

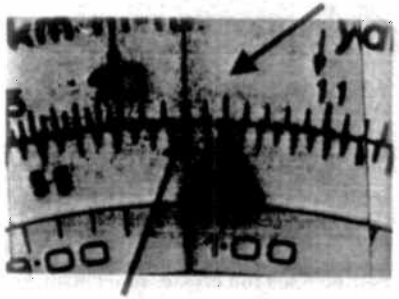
1. 60 min index aligned with 95 kts.

2. Cursor on 480 nm crosses inner scale at 303 min.

CRP5 98 a.
To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 105 kts on the outer scale.
b. Position the cursor over 103 nm on the outer scale and read off 59 min where cursor crosses the inner scale. This 00 hr 59 min (option a).
c. These settings are illustrated below.

1. 60 min index aligned with 105 kts.



2. Cursor on 103 nm crosses inner scale at 59 min.

CRP5 99 d.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 120 kts on the outer scale.
b. Position the cursor over 84 nm on the outer scale and read off 42 min where cursor crosses the inner scale. This 00 hr 42 min (option d).

CRP5 100 b.

To solve this type of problem using the CRP5:

a. Set 60 min index mark on inner scale against 135 kts on the outer scale.
b. Position the cursor over 433 nm on the outer scale and read off 192 min where cursor crosses the inner scale. This 3 hr 12 min (option b).
c. These settings are illustrated below.

1. 60 min index aligned with 135 kts.

2. Cursor on 433 nm crosses inner scale at 192 min.

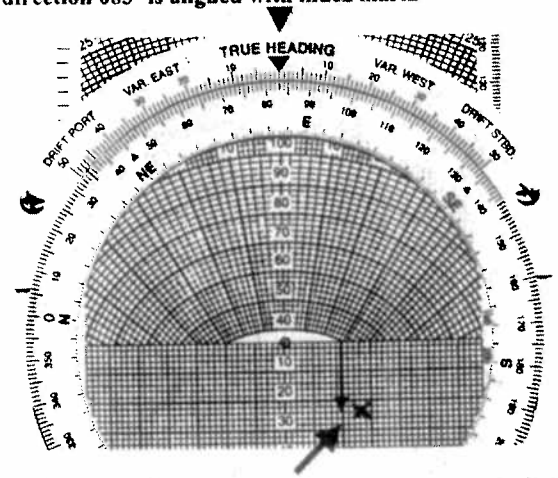
CRP5 101 a.

a. Set the low speed side of the slide so that the top line of the squared section aligns with the centre dot in the window.
b. Rotate the window to align the wind direction (035) with the true heading index mark.
c. Draw a cross to represent the wind speed (35 kts) on the centre line below the centre dot.

d. Rotate the window disc to align the runway direction (083) with the true heading index mark.

e. The headwind component (24 kts) (option a) is the vertical distance from the centre dot at the top of the gridded area, to the cross. This situation is illustrated below.

1. Runway direction 083° is aligned with index mark.



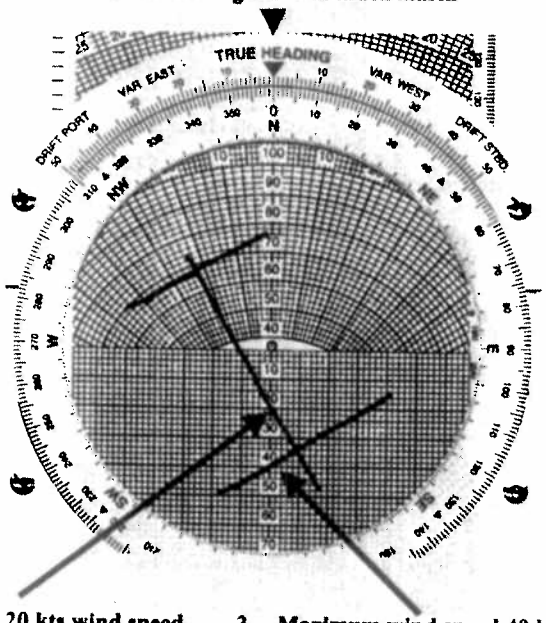
2. Vertical distance from dot at top of gridded area to cross indicates headwind component 24 kts.

CRP5 102 a.

To solve this type of problem using the CRP5:

a. Set the low speed side of the slide so that the top line of the squared section aligns with the centre dot in the window.
b. Rotate the window to align the wind direction (060) with the true heading index mark.
c. Draw a horizontal line 10 kts below the centre dot to represent the minimum headwind component.
d. Draw two vertical lines, one 35 kts to the left and one 35 kts to the right of the centre dot.
e. Rotate the window until 000 aligns with the true heading index mark.
f. Read off the minimum (20 kts) and maximum (40 kts) wind speeds where these lines cross the centre line below the centre dot. These values are option b. This situation is illustrated below.

1. 000° is aligned with index mark.



2. Minimum 20 kts wind speed indicated where this line crosses centre line.

3. Maximum wind speed 40 kts indicated where this line Crosses centre line.

CRP5 103 a.

To solve this type of problem using the CRP5:

a. Position the low speed slide so that the centre dot is on the top line of the square gridded area of the slide.
b. Rotate the window to align the wind direction 280° with the true heading index mark.
c. Draw a cross on the centre line 40 kts below the centre dot.
d. Rotate the window to align the runway direction 230° with the true heading index mark.
e. Read of the cross wind component be counting the squares between the center line and the cross 30 kts. This is closest to option a 31 kts.

CRP5 104 c.

To solve this type of problem using the CRP5:

a. Dividing 48 nm by 10 min gives a ground speed of 288 kts.
b. Set the high speed slide so that 275 kts TAS is at the centre dot.
c. Align 066° heading on the window scale with 11° E variation on the outer scale.

d. With a heading of 066° and a course of 075° the drift is 9° right. Draw a cross on the 288 kts arc at 9° right drift.
e. Rotate the window to align the cross on the centre line below the centre dot.
f. Read of wind direction 340° at the true heading index, and wind speed 45 kts by counting down from the centre dot to the cross. The wind is 340°/45 kts (option c).

CRP5 105 a.

To solve this type of problem using the CRP5:

a. Dividing 64 nm by 11 min gives a ground speed of 320 kts.
b. Set the high speed slide so that 360 kts TAS is at the centre dot.
c. Aiign 215° heading on the window scale with 15° E variation on the outer scale.
d. With a heading of 215° and a course of 210° the drift is 5° left. Draw a cross on the 320 kts arc at 5° left drift.
e. Rotate the window to align the cross on the centre line below the centre dot.
f. Read of wind direction 265° at the true heading index, and wind speed 50 kts by counting down from the centre dot to the cross. The wind is 265°/50 kts (option a).

CRP5 106 b.

It is not necessary to use the CRP5 to solve this type of problem. Drift angle is greatest when the wind is at 90° to the track. So for a track of 040° the greatest drift will occur when the wind is from 310° or 130° (option b).

CRP5 107 d.

To solve this type of problem using a CRP5:

a. Set temperature +20°C against altitude in thousands of feet (8) in the airspeed window.
b. With the cursor set at 120 kts on the inner scale it will cross the outer scale at 141 kts (option d).

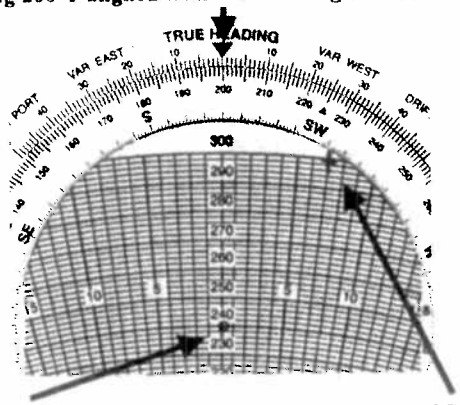
CRP5 108 d.

To solve this type of problem using the CRP5:

a. At 12000 ft in the ISA the temperature is 15°C – (12 x 2°C) = -9°C. To convert CAS to TAS set ISA conditions by aligning the –9°C with 12 in the airspeed window. With the cursor aligned with 200 kt on the inner scale it will cross the outer scale at 240 kts TAS.
b. Calculate ground speed by dividing the distance flown (105 nm) by the time taken 21 minutes then multiply by 60 to get 300 kts ground speed.
c. With heading 215°M and 15°W variation, the heading is 200°T.

d. With track 222°M and heading 215°M the drift is 7° right.
e. Set the low speed slide with TAS 240 kts at the centre dot.
f. Align heading 200°T with the true heading index.
g. Draw a cross to represent ground speed on the 300 kt radial at 7° right drift. This situation is illustrated below.

1. Heading 200°T aligned with true heading index mark.



2. Centre dot on 240 kts TAS.
3. Cross drawn at 300 kts GS, 7 right drift.

h. Rotate the window to align the cross on the centre line below the centre ot. Read off the 52° T/69 kts, or approximately 050°/70 kts (option d).

CRP5 109 b.
It is not necessary to use the CRP5 to solve this type of problem. Drift angle is greatest when the wind is at 90° to the course. So for a course of 095° the greatest drift will occur when the wind is from 005° or 185° (option b).

CRP5 110 c.
To solve this type of problem using the CRP5:

a. Set the 60 min index mark on the inners scale against 240 on the outer scale.
b. Against 500 on the outer scale read off 124.5 or approximately 125 min (option c) on the inner scale.

CRP5 111 c.
To solve this type of problem using the CRP5:

a. Flight time between X and Y = 1252 – 1232 = 20 min.
b. To find the ground speed divide the distance between X and Y, (40 nm) by time (20 min) by setting 20 on the inner scale against 40 0n the outer scale.

The 60 min index on the inner scale will now align with 120 on the outer. This means that the ground speed is 120 nm/hr.
c. To find the time taken to fly from Y to Z divide the distance (75 nm) by the ground speed (120). To do this, align 120 on the inner scale with 75 on the outer scale. The 60 min index on the inner will now align with 37.5 on the outer. This means that the time between Y and Z will be 37.5 min.
d. Adding this to the ATA at Y gives 1252 + 37.5 min = 1329.5 or approximately 1330 UTC (option c).

CRP5 112 d.
To solve this type of problem using the CRP5:

a. Flight time between A and B = 1027 – 1010 = 17 min.
b. To find the ground speed divide the distance between A and B, (30 nm) by time (17 min) by setting 17 on the inner scale against 30 on the outer scale. The 60 min index on the inner scale will now align with 105.8 on the outer. This means that the ground speed is 105.8 nm/hr.
c. To find the time taken to fly from B to C divide the distance (20 nm) by the ground speed (105.8). To do this, align 105.8 on the inner scale with 20 on the outer scale. The 60 min index on the inner will now align with 11.35 min on the outer. This means that the time between B and C will be 11.35 min.
d. Adding this to the ATA at C gives 1027 + 11.35 min = 1038.25 or approximately 1038 UTC (option d).

CRP5 113 b.
To solve this type of problem using the CRP5:

a. Flight time between X and Y = 1450 – 1420 = 30 min.
b. To find the ground speed divide the distance between A and B, (30 nm) by time (30 min) by setting 30 on the inner scale against 30 on the outer scale. The 60 min index on the inner scale will now align with 60 on the outer. This means that the ground speed is 60 nm/hr.
c. To find the time taken to fly from Y to Z divide the distance (20 nm) by the ground speed (60). To do this, align 60 on the inner scale with 20 on the outer scale. The 60 min index on the inner will now align with 20 min on the outer. This means that the time between B and C will be 20 min.
d. Adding this to the ATA at Y gives 1450 + 20 min = 1510 UTC (option b).

CRP5 114 d.
To solve this type of problem using the CRP5:

a. To set the CRP5 to ISA conditions, set 10 on the inner scale and 10 on the outer scale.
b. Opposite 5000 ft in the altitude window set 35°C.
c. Opposite 5 (to represent 5000) on the inner scale read off 5550 on the outer scale. This means that the true altitude is 5550 ft (option d).

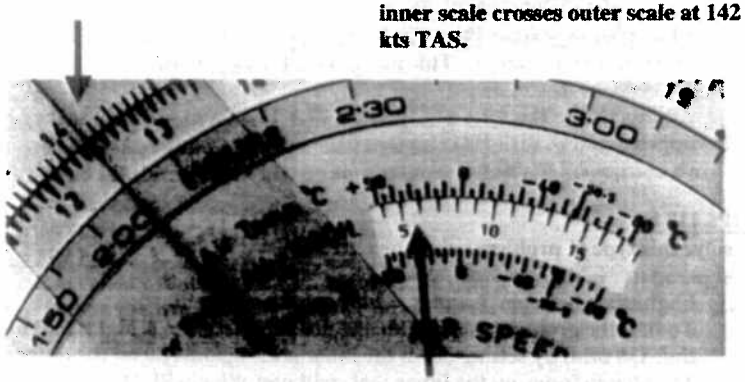
CRP5 115 a.
To solve this type of problem using the CRP5:

a. **To set the CRP5 to ISA conditions, set 10 on the inner scale and 10 on the outer scale.**
b. **Opposite 7000 ft in the altitude window set 30°C.**
b. **Opposite 7 (to represent 7000) on the inner scale read off 7750 on the outer scale. This means that the true altitude is 7750 ft (option a).**

CRP5 116 a.
To solve this type of problem using a CRP5:

a. **Set temperature +25°C against altitude in thousands of feet (6) in the airspeed window.**
b. **With the cursor set at 125 kts on the inner scale it will cross the outer scale at 142 kts (option a). These settings are illustrated below.**

2. Cursor aligned with 125 kts on inner scale crosses outer scale at 142 kts TAS.

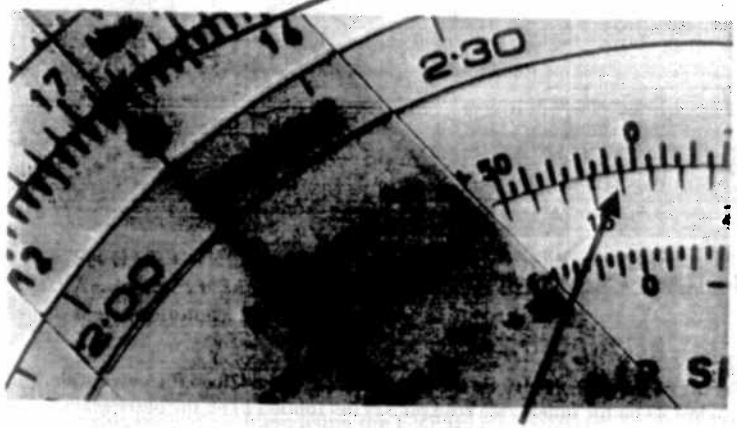


1. 25°C set against 6000 ft in Airspeed window.

CRP5 117 b.
To solve this type of problem using a CRP5:

a. **Set temperature +5°C against altitude in thousands of feet (16) in the airspeed window.**
b. **With the cursor set at 130 kts on the inner scale it will cross the outer scale at 173 kts (option b). These settings are illustrated below.**

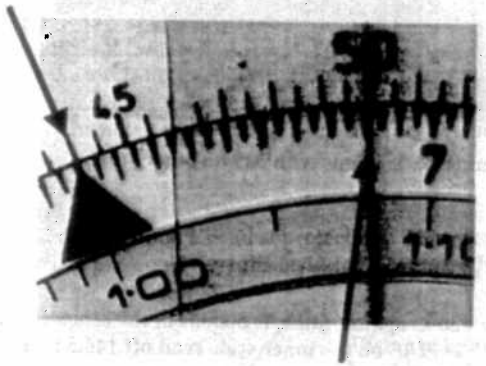
2. Cursor allgned with 130 kts on inner scale crosses outer scale at 173 kts TAS.



1. 5°C set against 16000 ft in Airspeed window.

CRP5 118 a.
a. **Set the 60 min index mark on the inners scale against 440 on the outer scale.**
b. **Against 500 on the outer scale read off 68.2 or approximately 68 min (option a) on the inner scale. These settings are illustrated below.**

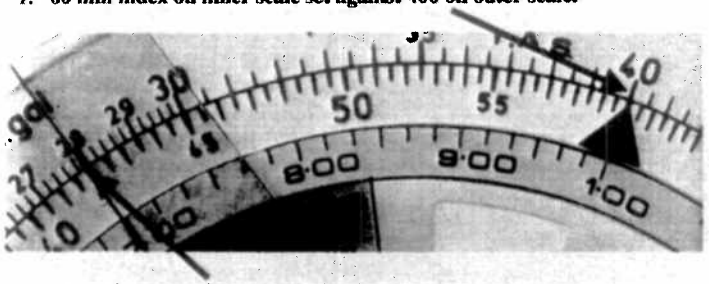
1. 60 min index on inner scale set against 440 on outer scale.



2. 500 on outer scale aligns with 68.2 on inner scale.

CRP5 119 c.
a. **Set the 60 min index mark on the inners scale against 400 on the outer scale.**
b. **Against 42 on the inner scale read off 280 nm (option c) on the outer scale. These settings are illustrated below.**

1. 60 min index on inner scale set against 400 on outer scale.



2. 42 on inner scale aligns with 280 on outer scale.

CRP5 120 b.

a. Set the 60 min index mark on the inners scale against 420 on the outer scale.
b. Against 47 on the inner scale read off 329 nm (option b) on the outer scale. These settings are illustrated below.

1. 60 min index on inner scale set against 420 on outer scale.

2. 47 on inner scale aligns with 329 on outer scale.

CRP5 121 d.

To solve this type of problem using the CRP5:

a. Set OAT = +20°C against altitude 6000 ft in the airspeed window.
b. Against 130 kts IAS on the inner scale read off 146.5 kt TAS on the outer scale (option d).

CRP5 122.

To solve this type of problem using the CRP5:

a. Set OAT = +25°C against altitude 2000 ft in the airspeed window.
b. Against 110 kts IAS on the inner scale read off 116 kt TAS on the outer scale (option a).

CRP5 123 a.

To solve this type of problem using the CRP5:

a. Set OAT = +15°C against altitude 4500 ft in the airspeed window.
b. Against 100 kts IAS on the inner scale read off 100 kt TAS on the outer scale (option a).

CRP5 124 d.

To solve this type of problem using the CRP5:

a. Set OAT = -15°C against altitude 7500 ft in the airspeed window.
b. Against 97 kts IAS on the inner scale read off 105.5 kt TAS on the outer scale (option d).

CRP5 125 a.

To solve this type of problem using the CRP5:

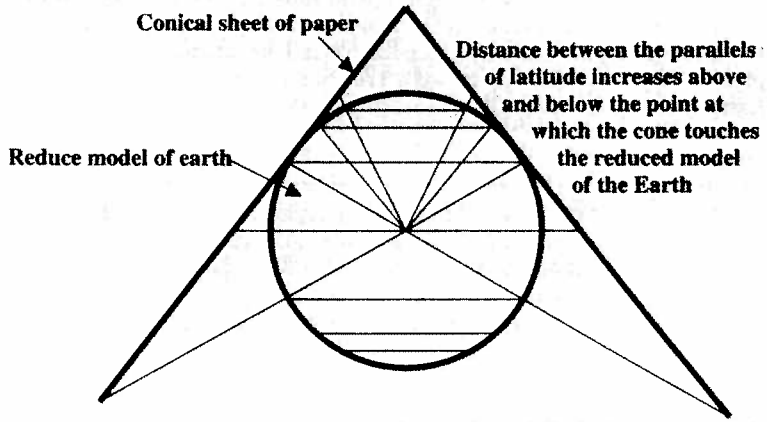
c. Set OAT = +10°C against altitude 2500 ft in the airspeed window.
d. Against 89 kts IAS on the inner scale read off 92.5 kt TAS on the outer scale (option a).

CHART THEORY 1 a.

The terms "magnetic variation" refers to the difference between True North and Magnetic North directions at any point on the earth. Variation is caused by the fact that the Magnetic and Geographical North Poles are not in exactly the same place. Variation is the angle between True North and Magnetic North, so a Westerly variation means that true North is to the East of magnetic North (option a). It should be noted that options c and d refers of compass deviation.

CHART THEORY 2 a.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. The nominal scale of a Lambert's Conformal chart is therefore correct only along the standard parallels (option a).

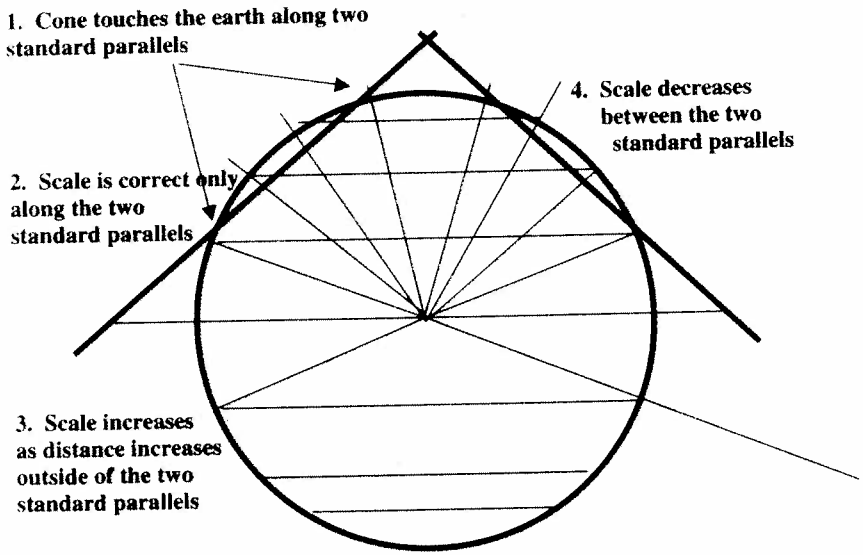


CHART THEORY 3 b.

The terms "magnetic variation" refers to the difference between True North and Magnetic North directions at any point on the earth. Variation is caused by the fact that the Magnetic and Geographical North Poles are not in exactly the same place. Variation is the angle between True North and Magnetic North, so an Easterly variation means that true North is to the West of magnetic North

CHART THEORY 4 c.

The most suitable chart for use in any given region of the world is that which gives the most accurate representation of sizes and relative positions of the various features within that region. In the case of the polar regions, the most accurate representation is provided by a Polar Stereographic Chart (option c). This is produced by projecting the graticule of the reduce Earth onto a flat sheet of paper that is in tangential to the reduced Earth at the pole as illustrated

below. The scale of such a chart is most accurate in the region very close to the pole, but becomes increasing inaccurate as distance from the pole increases.

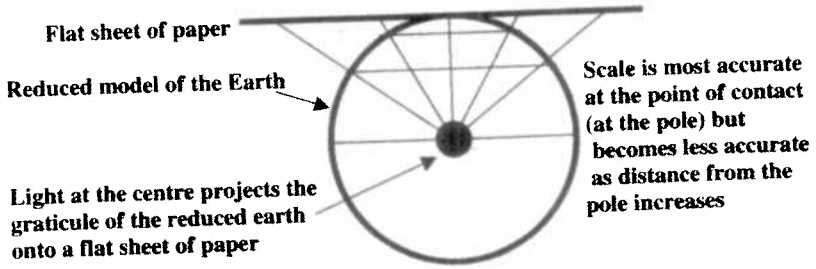
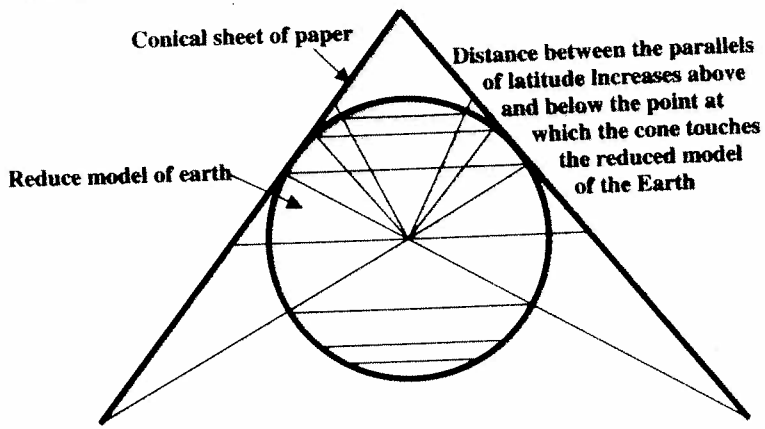


CHART THEORY 5 a.

The term "chart convergency" refers to the angle of inclination between two meridians on the chart measured at any latitude. A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale and chart convergency to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale and chart convergency can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. Scale on a Lambert's Conformal chart are therefore most accurate along the standard parallels. But the convergence of the meridians is most accurately represented at the parallel of origin, which is located midway between the two standard parallels (option a).

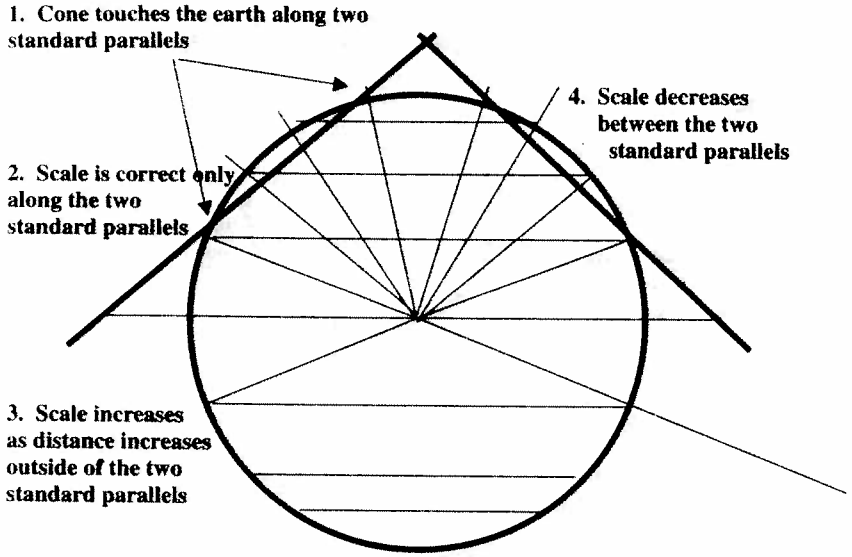
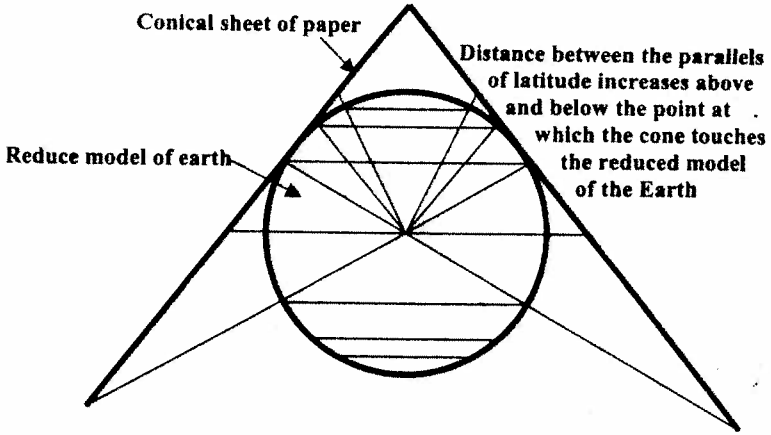


CHART THEORY 6 c.

A Rhumb Line is a regularly curved line on the surface of the earth that cuts all meridians in its path at the same angle. Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the model to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians are straight lines. This means that a straight-line track drawn on the chart will cross al meridians at the same angle. A Rhumb line track on a Mercator chart is therefore a straight line (option c).

CHART THEORY 7 c.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. The nominal scale of a Lambert's Conformal chart is therefore correct only along the standard parallels (option c).

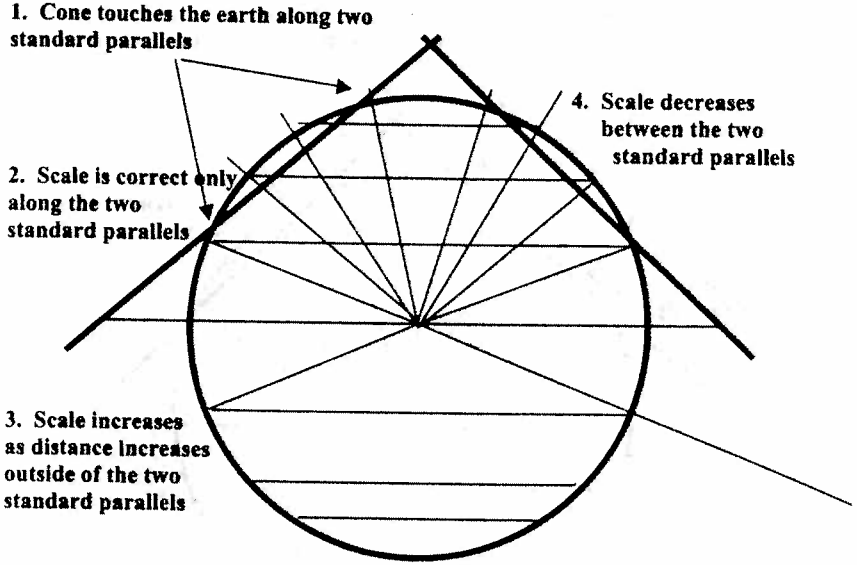
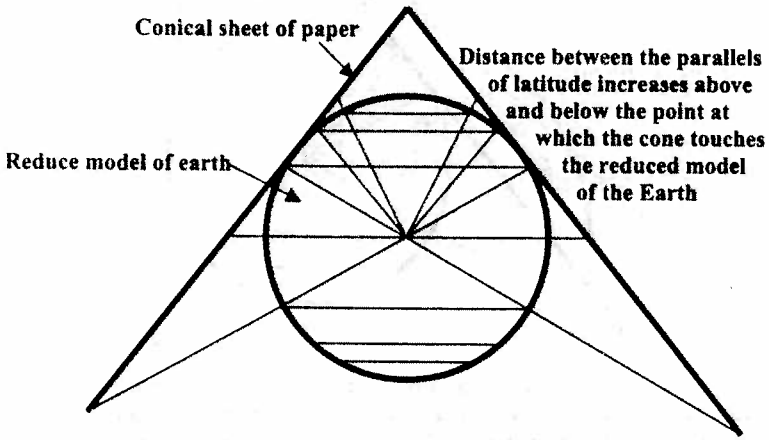


CHART THEORY 8 b.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. The nominal scale of a Lambert's Conformal chart is therefore correct only along the standard parallels (option b).

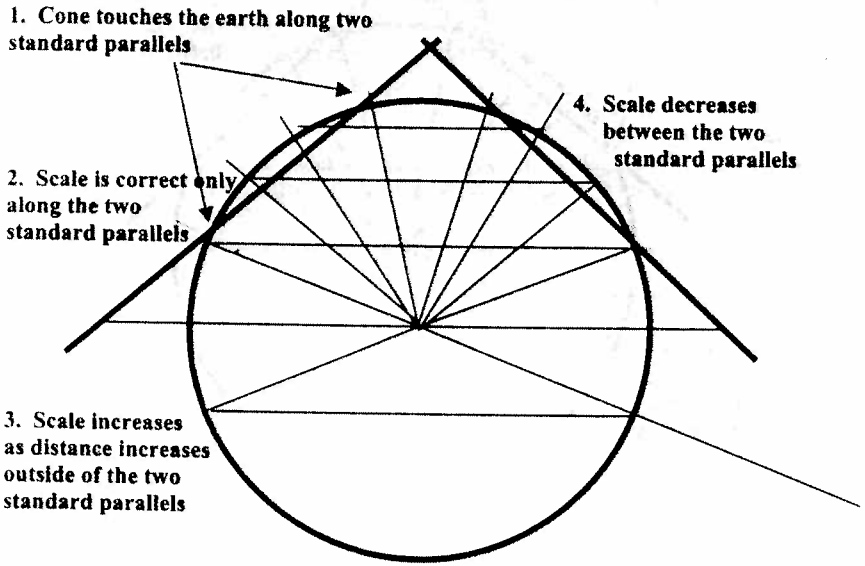


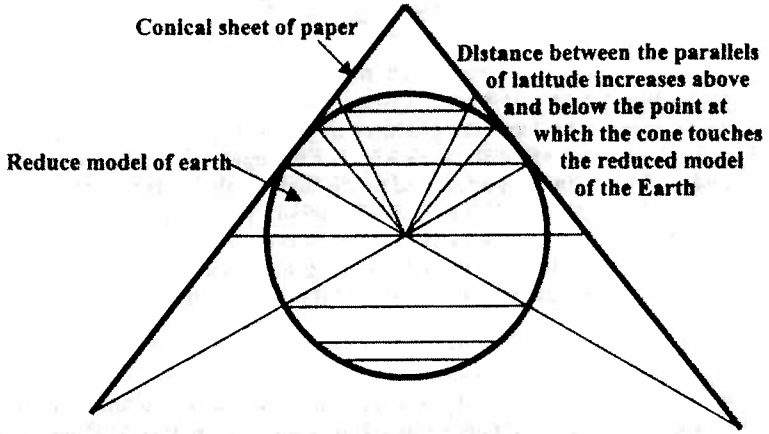
CHART THEORY 9 c.

A Great Circle is a circle drawn on the surface of the earth such that its centre is the centre of the earth. Great circles include all meridians plus the equator. Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the equator to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians and parallels of latitude, including the Equator are shown as straight lines. Great circle representing the meridians and the equator are therefore shown as straight lines.

But Great Circles may also be inclined at any angle, provided their centre is located at the centre of the earth. Such Great Circles will cross each meridian or parallel of latitude in their paths at slightly different angle to the one before. This means that with the exception of the Equator and the meridians, great circles are shown as curves concave to the equator on Direct Mercator charts (option c).

CHART THEORY 10 d.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. The nominal scale of a Lambert's Conformal chart is therefore correct only along the standard parallels. But

because the scale increases at either side of the parallel of origin, the convergence is most accurate at the parallel of origin, which is located midway between the standard parallels (option d).

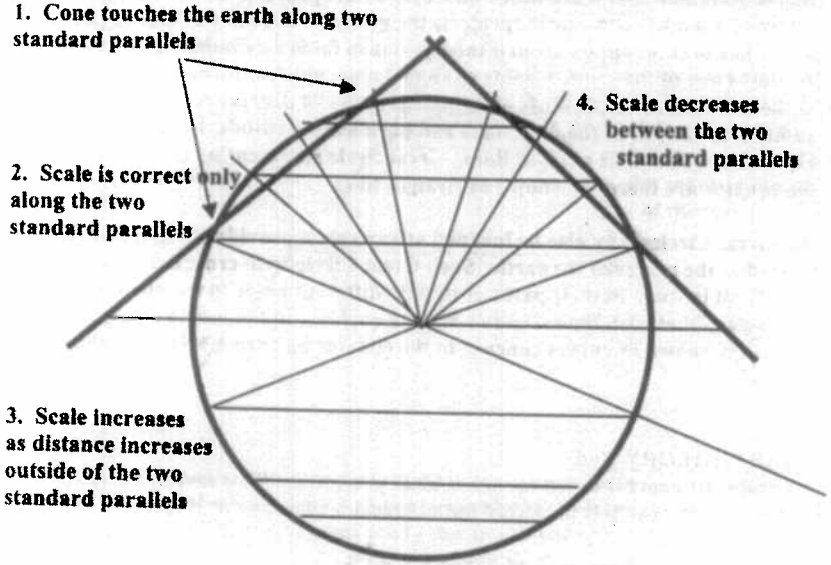


CHART THEORY 11 a.

Transverse Mercator charts are produced by projecting the graticule of a model of the earth onto the inside of a sheet of paper wrapped around the earth to form a cylinder. This cylinder is aligned so that it contacts the surface of the earth at the poles and along a meridian and anti-meridian. This projection is then mathematically adjusted to achieve orthomorphism. The scale on such a chart is exactly correct only along the meridian of tangency, which is the line along which the cylinder is in contact with the surface of the earth (option a).

CHART THEORY 12 b.

A Lamberts chart is produced by projecting the graticule of a model of the earth onto the inside of a cone of paper which is then unfolded to form the chart. This process produces a graticule in which the meridians are straight lines converging upon the nearest pole. In the Northern hemisphere each meridian points directly at the North Pole. Because the meridians converge, any straight line not in a north-south direction on such a chart will cross each successive meridian in its path at a slightly different angle. This means that the direction of the track will change continuously along any straight line track. This change of track is caused by the convergence of the meridians, so the difference in the departure and arrival track between two points is equal to the convergence of the chart (option b). This situation is illustrated below.

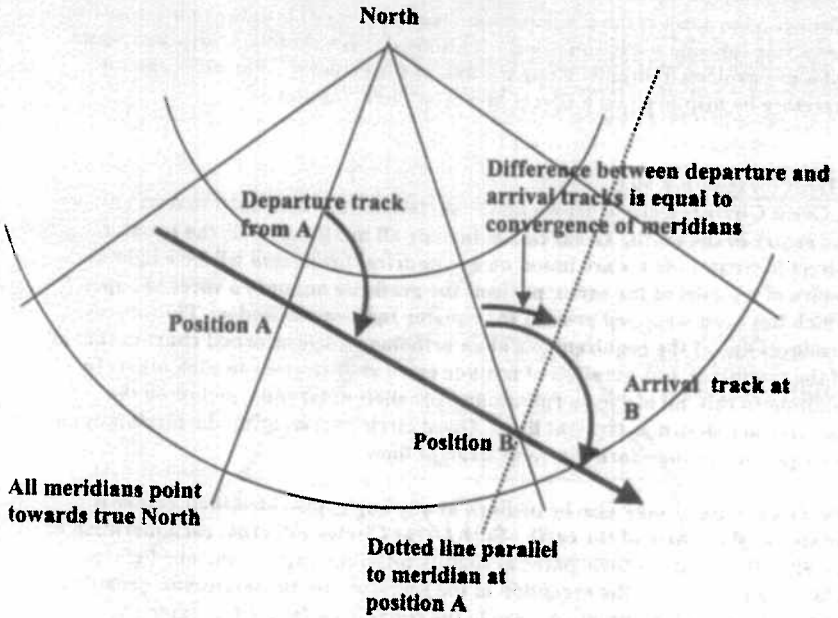


CHART THEORY 13 a.

A Rhumb Line is a regularly curved line on the surface of the earth that cuts al of the meridians in its path at the same angle. A second property of Rhumb Lines is that when extended to circle the earth, the centre of this circle is not located at the centre of the earth. This in turn means that the r5adius of curvature of a Rhumb Line is less than that of the earth. Because all of the parallels of latitude cross all meridians at the same angle, they are all Rhumb Lines, with the exception of the Equator (option a).

CHART THEORY 14 c.

Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the model to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians are straight lines. This means that all meridians on a Direct Mercator chart are parallel, equally spaced, vertical straight lines (option c).

CHART THEORY 15 d.

Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the equator of the model to form a cylinder.

This process causes the spacing between the parallels of latitude to increases with increasing latitude north and south. This also means that the North and South poles are projected directly along the axis of the cylinder. The poles cannot therefore be displayed on a Direct Mercator chart (option d).

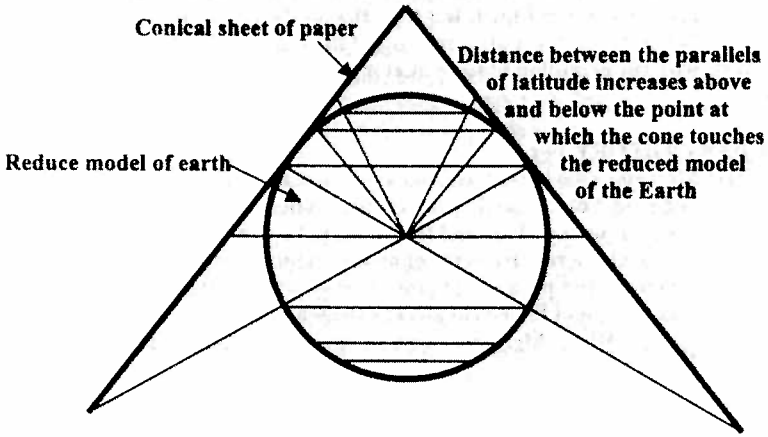
CHART THEORY 16 a.

A Great Circle is a circle drawn on the surface of the earth such that its centre is the centre of the earth. Great circles include all meridians plus the equator. Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the equator to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians and parallels of latitude, including the Equator are shown as straight lines. Great circle representing the meridians and the equator are therefore shown as straight lines.

But Great Circles may also be inclined at any angle, provided their centre is located at the centre of the earth. Such Great Circles will cross each meridian or parallel of latitude in their paths at slightly different angle to the one before. This means that with the exception of the Equator and the meridians, great circles are shown as curves concave to the equator on Direct Mercator charts (option a).

CHART THEORY 17 b.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points.

This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. This means that the distance between parallels of latitude equal numbers of degrees apart, reduces between and increases outside of the standard parallels (option b).

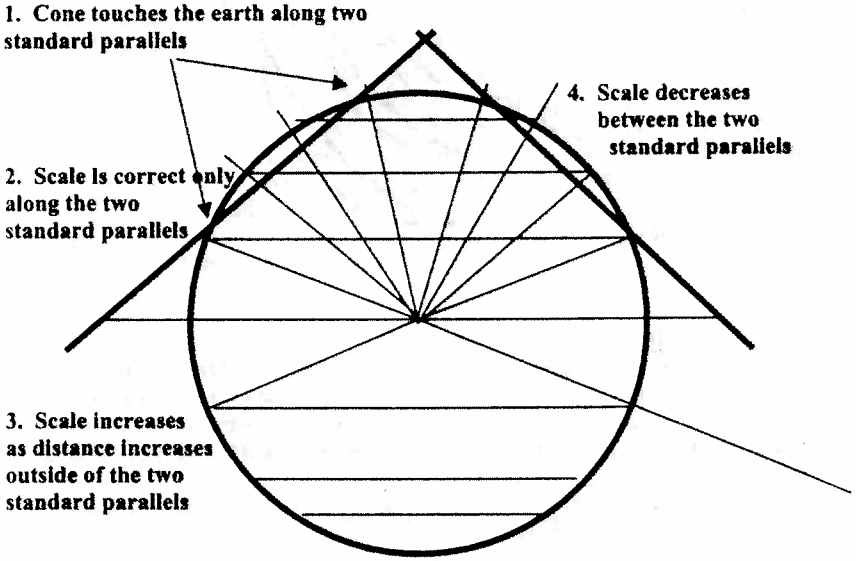


CHART THEORY 18 a.

A Polar Stereographic chart is produced by projecting the graticule of a model of the Earth onto a flat sheet of paper that is in contact with the model at the pole. This produces a series of concentric circles centred on the pole. On all such projections Rhumb Lines and Great Circles appear as curves, such that the Great Circle is always closer to the pole than is the Rhumb Line. But as illustrated below, as latitude increases towards the pole, the curvature of the Great Circle gradually reduces. This means that the higher the latitude, the closer the Great Circle approximates to a straight line (option a).

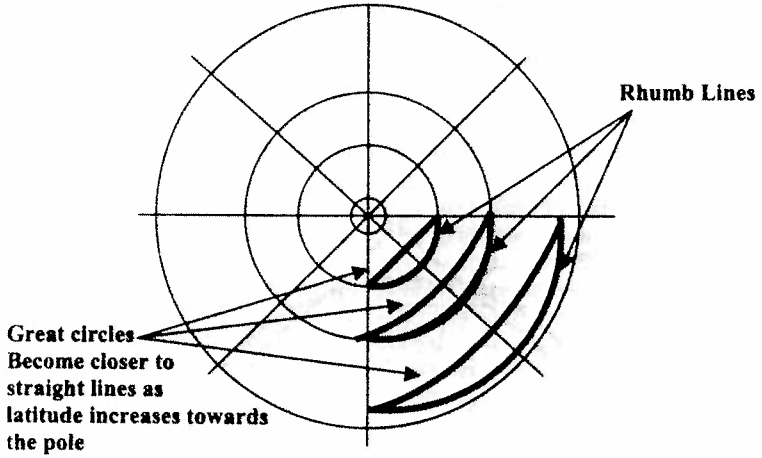


CHART THEORY 19 b.

A Polar Stereographic chart is produced by projecting the graticule of a model of the Earth onto a flat sheet of paper that is in contact with the model at the pole. This produces a series of concentric circles centred on the pole. On all such projections Rhumb Lines and Great Circles appear as curves concave to the pole (option b).

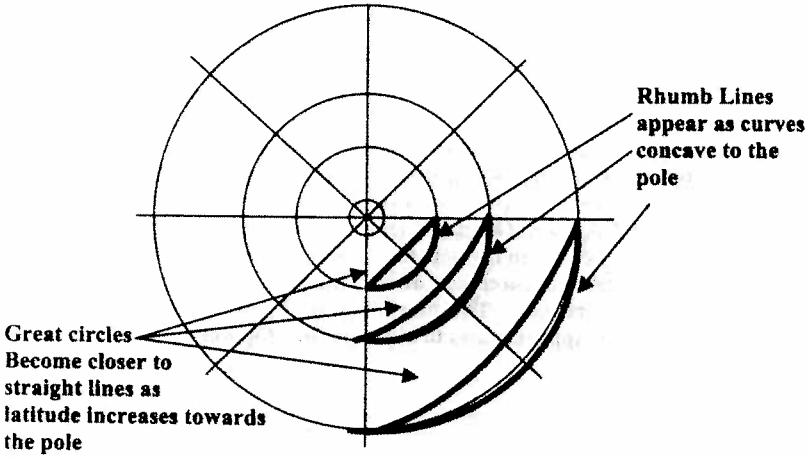


CHART THEORY 20 c.

A Polar Stereographic chart is produced by projecting the graticule of a model of the Earth onto a flat sheet of paper that is in contact with the model at one of the poles. This produces a series of concentric circles centred on the pole and the meridians are represented as straight lines radiating out from the pole. The

scale of a polar stereographic is correct only at the pole and increases with increasing distance from the pole. The term "Convergence factor" refers to the ratio of chart convergence to earth convergence and is 1 when these two convergences are equal. On a polar stereographic chart this convergence factor 1 only at the pole (option c).

CHART THEORY 21 a.

A Rhumb Line is a regularly curved line on the surface of the earth that cuts all meridians in its path at the same angle. Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the model to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians are straight lines. This means that a straight-line track drawn on the chart will cross al meridians at the same angle. A Rhumb line track on a Mercator chart is therefore a straight line (option a).

CHART THEORY 22 c.

A Lamberts chart is produced by projecting the graticule of a model of the earth onto the inside of a cone of paper which is then unfolded to form the chart. This process produces a graticule in which the meridians are straight lines converging upon the nearest pole. The scale of such a projection is correct only along the line at which the cone touches the model of the earth. To improve the accuracy of the chart it is mathematically adjusted to simulate the situation in which the cone passes slightly below the surface of the model earth. This produces two standard parallels along which the scale is correct. These standard parallels are the lines along which the cone would pass through the surface of the model earth. The parallel midway between the two standard parallels is called the parallel of origin. The graticule produced by this process is illustrated below.

The only truly straight lines on such a chart are the parallels of longitude (option c). Great circles represented by curve concave to the parallel of origin and are approximately straight line when very close to it. But the only perfectly straight lines on such a chart are the meridians of latitude.

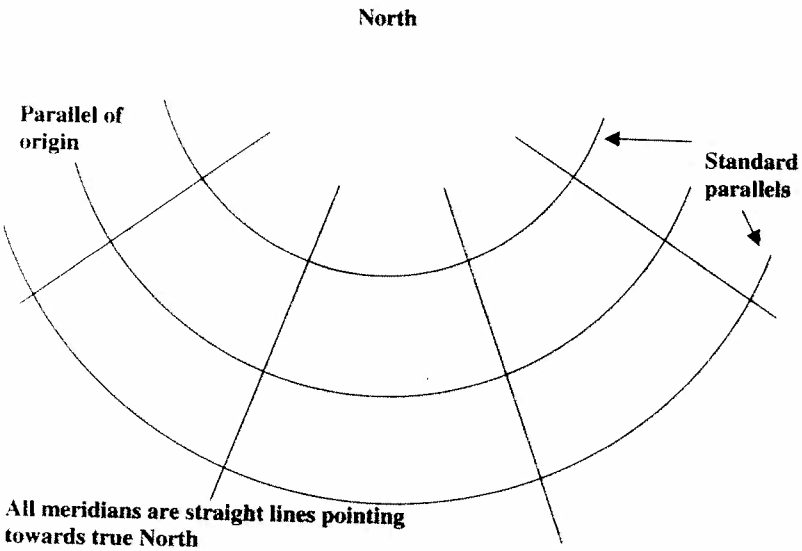


CHART THEORY 23 d.

Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the model to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians are straight lines. But the scale on such charts is correct only at the Equator, then expands directly with the secant of the latitude north and south (option d).

CHART THEORY 24 b.

A rhumb line is part of a circle around the earth, which does not have its centre at the centre of the earth. This means that the radius of a rhumb line circle is always less than that of the earth. It is possible for rhumb lines to be oriented in any direction, so their appearance on any chart will depend upon the direction from which the earth is being viewed and the direction of the rhumb line. This can be demonstrated by considering any parallel of latitude. When viewed from a point at the same latitude as the rhumb line it will appear to be a straight line. But when viewed from one of the poles it will appear to be a circle or part of a circle with its centre at the pole. A Polar stereographic chart represents the earth as viewed from above one of the poles. This means that any rhumb line other than those running directly north-south, will appear as curves concave to the nearer pole (option b).

CHART THEORY 25 a.

Transverse Mercator charts are produced by projecting the graticule of a model of the earth onto the inside of a sheet of paper wrapped around the earth to form a cylinder. This cylinder is aligned so that it contacts the surface of the earth at the poles and along a meridian and anti-meridian. This projection is then mathematically adjusted to achieve orthomorphism. The line of contact between the cylinder and the earth is called the meridian of tangency, and this is the only meridian that appears a straight line on the chart. All of the other meridians are distorted such that they appear as complex curves. Because of this distortion, the convergency of the chart is correct only along the datum meridian and at the poles and equator (option a).

CHART THEORY 26 d.

A Polar Stereographic Chart (option d) is produced by projecting the graticule of the reduce Earth onto a flat sheet of paper that is in tangential to the reduced Earth at the pole as illustrated below.

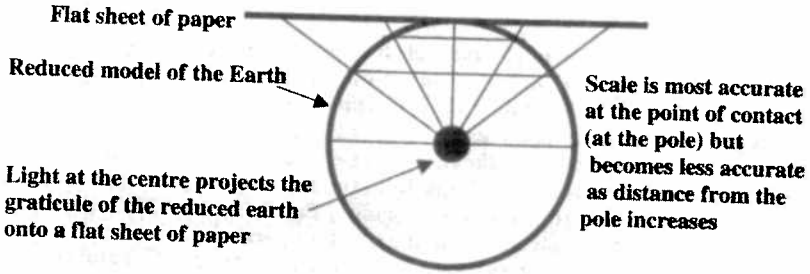


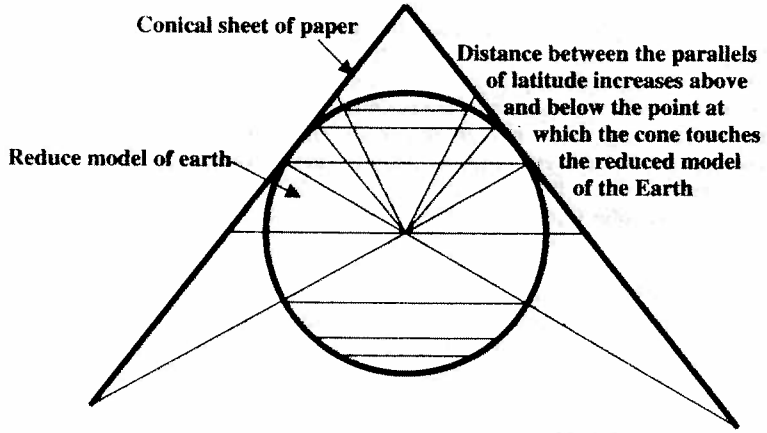
CHART THEORY 27 b.

A Great Circle is a circle drawn on the surface of the earth such that its centre is the centre of the earth. Great circles include all meridians plus the equator. Direct Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the equator to form a cylinder. This process produces one of the requirements of an orthomorphic/conformal chart in that all of the meridians and parallels of latitude cross at 90 degrees to each other. In addition to this, all of the meridians and parallels of latitude, including the Equator are shown as straight lines. Great circle representing the meridians and the equator are therefore shown as straight lines.

But Great Circles may also be inclined at any angle, provided their centre is located at the centre of the earth. Such Great Circles will cross each meridian or parallel of latitude in their paths at slightly different angle to the one before. This means that with the exception of the Equator and the meridians, great circles are shown as curves convex to the nearest pole on Direct Mercator charts (option b).

CHART THEORY 28 b.

A Lambert's chart is based upon a conical projection. This means that the graticule of the earth is projected onto the inside of a cone as illustrated below.



This method of projection causes the scale to increase as distance increases away from the line at which the cone is in contact with the model of the Earth. The scale can therefore be correct only along this line of contact. In order to make such a projection conformal the scale is reduced mathematically at all points. This is the equivalent of having the cone pass below the surface of the model of the earth as illustrated below. This produces two lines of parallel at which the scale is correct. These are termed the "Standard Parallels". Scale between these parallels reduces while scale outside of the standard parallels increases as distance away from the parallels increases. The nominal scale of a Lambert's Conformal chart is therefore constant along any parallel of latitude (option b).

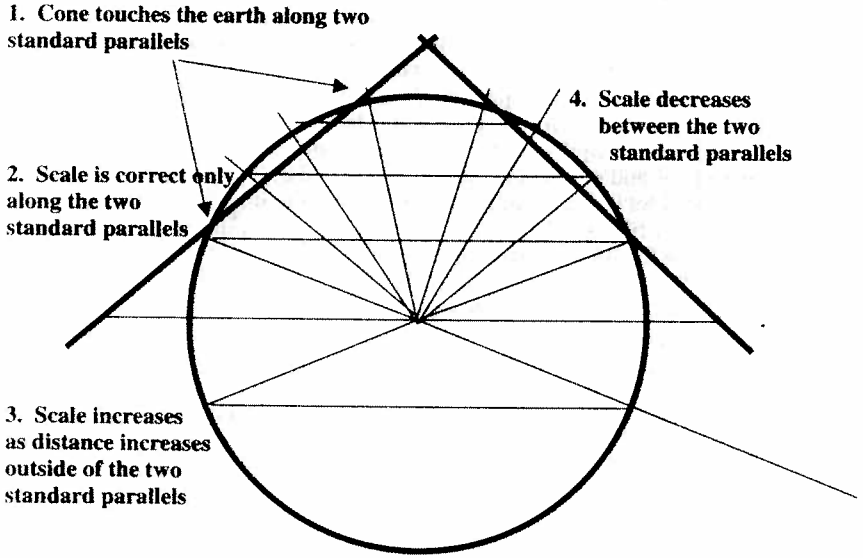


CHART THEORY 29 c.

A Great Circle is a circle drawn on the surface of the earth such that its centre is the centre of the earth. Great circles include all meridians plus the equator. Mercator charts are based on a cylindrical projection where a light at the centre of a model of the earth, projects the graticule out onto a sheet of paper, which has been wrapped around the earth to form a cylinder. In the case of Transverse Mercator charts the line of contact between the paper cylinder and the earth is one of the meridians. This meridian is called the great circle of tangency. One of the effects of this form of projection is that the scale is correct only along this great circle of tangency (option c) then expands with increasing distance either side.

CHART THEORY 30.

A flight Information region (FIR) boundary is indicated by a line _ _ _ _ _ (option b).

CHART THEORY 31.

The symbol for an uncontrolled route is marked with a - line (option d).

CHART THEORY 32.

The symbol for a Control Zone (CTR) on an aeronautical chart i option a.

— — — — — —

CHART THEORY 33 b.

Symbol 8 is a lightship, 6 is an exceptionally high obstacle, 13 is a VOR, 15 is an unlighted obstacle.

CHART THEORY 34 a.

Symbol 13 is a VOR, 8 is a lightship, 6 is an exceptionally high obstacle, 15 is an unlighted obstacle.

CHART THEORY 35 b.

Symbol 15 is an unlighted obstacle, 1 is a basic radio aid, 13 is a VOR, and 14 is a VOR TAC.

CHART THEORY 36 c.

Symbol 3 is an aeronautical ground light, 10 is an exceptionally high lighted obstacle, 14 is a VOR TAC, and 15 is an unlighted obstacle.

CHART THEORY 37 d.

Symbol 14 is a VOR TAC, 1 is a basic radio aid, 3 is an aeronautical light and 7 is a VOR DME.

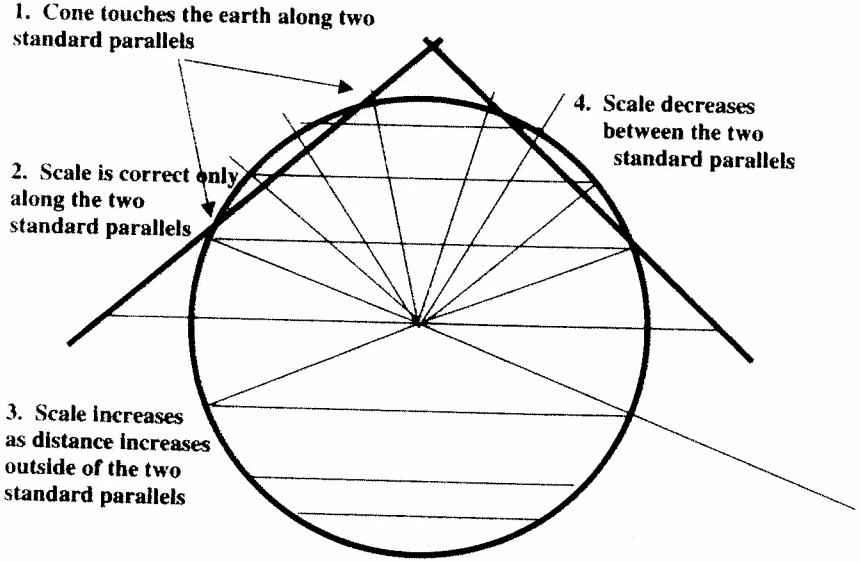
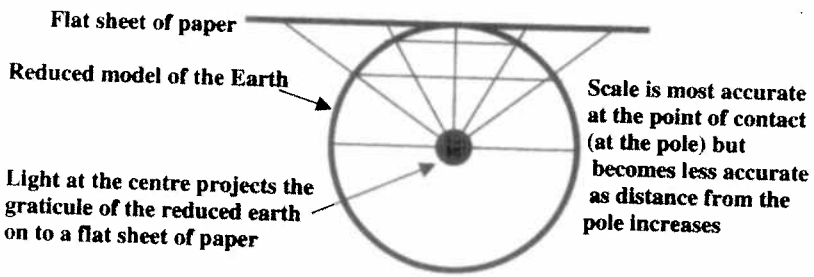


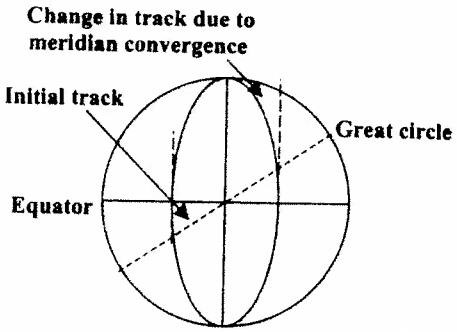
CHART THEORY 50 b.

The most suitable chart for use in any given region of the world is that which gives the most accurate representation of sizes and relative positions of the various features within that region. In the case of the polar regions, the most accurate representation is provided by a Polar Stereographic Chart (option b). This is produced by projecting the graticule of the reduce Earth onto a flat sheet of paper that is in tangential to the reduced Earth at the pole as illustrated below. The scale of such a chart is most accurate in the region very close to the pole, but becomes increasing inaccurate as distance from the pole increases.



CIRCLES 1 a.

The track at any point on a great circle is the angle between the great circle direction and the meridian at that point. All meridians are parallel at the equator, but gradually converge as latitude increases towards either of the two poles. This causes the track of great circles other than the meridians and the equator, to change as latitude changes. This effect is illustrated at the right.



This change of track between any two points on a great circle is equal to the convergency of the meridians between these two points. The change in great circle track between two points can be calculated using the convergency equation below:

Convergency = change in longitude x Sin of mean latitude.

The change in longitude = initial longitude – final longitude

Which in this question = 148^0 W – 141^0 W = 7^0

The mean latitude = (initial latitude + final latitude) / 2

Which in this question = (59^0 S + 61^0 S) / 2 = 60^0

Inserting these figures into the standard equation gives:

Convergency = 7^0 x sin 60 = 6.06^0

So between points A and B, the great circle track increases by approximately 6^0 (option a).

CIRCLES 2 a.

The distance in nm between any two meridians along a specified parallel of latitude is called the departure. This can calculated using the departure equation:

Departure (in nm) = Change in longitude x Cos latitude

This can be rearranged to give:

Change in longitude = Departure (in nm) / Cos latitude.

In this question the latitude is constant at 45^0 N, and the departure is 280 nm.

Inserting these figures into the above equation gives:

Change in longitude = 280 nm/Cos 45° = 395.98' which divided by 60 = is 6° 36'.

The question states that point B is to the east of point A so subtracting the above figure from the longitude of b gives: B (45° 15'E) – (6° 36') = 38° 39' (option a).

CIRCLES 3 a.

To solve this type of problem the following points must be noted:

a. A straight line on a Lambert conformal chart is a great circle.
b. The direction of great circles changes along their lengths by an amount equal to the convergence of the meriadians between any two points on the great circle.
c. For a Lambert conformal chart the equation for the convergence of meridians is:

Convergence = Change in longitude x constant of the cone

The initial course at point A (40° N 050° W) is 043° (T) and at point B is 055° (T). This means that the difference in course is 12°, which is also the convergence between the meridians at A and B. So the convergence between points A and B is 12°.

The question states that the constant of the cone is 0.75.

Inserting these values into the convergence equation above gives:

12° = Change in longitude x 0.75

Rearranging this gives Change in longitude = 12° / 0.75 = 16°.

The initial longitude at point A is 050° W so the longitude at point B is 050° W - 16° = 34° (option a).

CIRCLES 4 c.

The direction of a great circle track varies along its length such that it cuts each meridian a slightly different angle. These changes are caused by the convergence of the meridians as latitude increases towards either of the two poles. The difference between the great circle direction and the rhumb line direction is the conversion angle. This angle can be calculated using the standard equation:

Conversion angle = ½ the change in longitude x the sin of the mean latitude

The change of longitude is the difference between the final and initial longitudes, which in this question is from 030°W, to 060°E = 90 degrees.

The latitude is constant at 70°S so the conversion angle in this question is:

Conversion angle = ½ x 90° x sin 70° = 42.29°.

The latitude is constant so the rhumb line direction is 90° so the great circle direction is 90° + 42.29° = 132.29° or approximately 132° (option c).

CIRCLES 5 c.

The distance in nm between any two meridians along a specified parallel of latitude is called the departure. This can calculated using the departure equation:

Departure (in nm) = Change in longitude x Cos latitude

This can be rearranged to give:

Change in longitude = Departure (in nm) / Cos latitude.

In this question the latitude is constant at 58° 42' N, and the departure is 6 nm.

Inserting these figures into the above equation gives:

Change in longitude = 6 nm / Cos 58° 42' = 11.55'.

The question states that point B is to the east of point A so subtracting the above figure from the longitude of A gives:

Longitude of B = (094° 00'W) – (11.55') = 93° 48.45' W (closest to option c).

CIRCLES 6 b.

This question states that an incorrect longitude has been put into the INS. An INS system will accept such an error, and will calculate all subsequent longitude positions based upon the error. This means that all subsequent longitudes with include the same magnitude of error as the longitude originally put into the system.

The initial error is the difference between the longitude put into the INS and the actual longitude. If the two longitudes are in different directions the error can be calculated using the equation:

Longitude error = input longitude + actual longitude

Inserting the data provided gives:

Longitude error = (139° 46.3'W) + (139° 46.3'E) = 279° 32.6' The input position is to the west of the actual position so the error is 279° 32.6' W

When the aircraft subsequently arrives at position point 52° N 180°W, the indicated position will be 279° 32.6' W of that point.

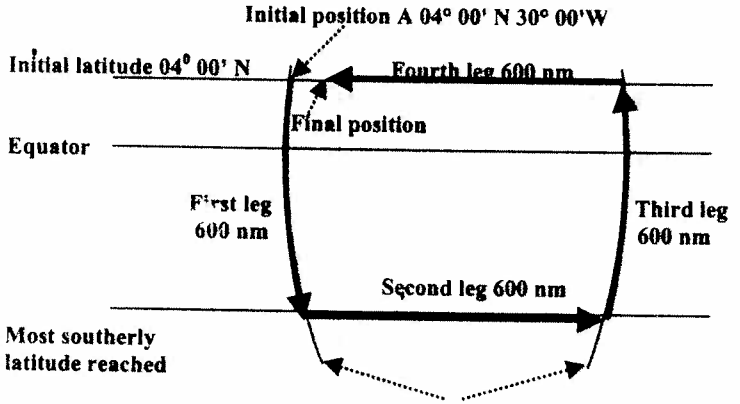
So the indicated position will be

(52° N 180°W) + (279° 32.6' W) = 459° 32.6' W.

But longitudes must be within the range 180° E to 180° W so by subtracting 360°, the indicated position of 459° 32.6' W is more properly expressed as 99° 32.6' W, which is closest to option b.

CIRCLES 7 c.

When solving this type of question it is helpful to start with a sketch of the situation as illustrated below.



Meridian converge to the North and South of the equator

To calculate the change in latitude during the first leg (Northerly) and third leg (Southerly) use the fact that each nm along any meridian is equal to 1' latitude. So 600 nm = 600' which is 10° change of latitude.

So the most Southerly latitude reached is 10° South of the initial latitude of (04° 00' N). The most Southerly latitude is therefore 06° 00' S.

During the second leg the aircraft flies 600 nm to the East at latitude 06° 00' S. Using the departure equation to calculate the change in longitude gives:

Departure = Change in longitude x Cos latitude

Which can be rearranged to give: Change in longitude = Departure/Cos latitude

Inserting the data above gives:

Change in longitude = 600 nm / Cos 06° 00' which = 603.3' E.

Dividing this by 60 gives change in longitude = 10° 3.3' E so the longitude at the end of the second leg:

Longitude = initial longitude (030° 00'W) + (10° 3.3' E) = 19° 56.7' W

During the fourth leg at latitude 04° 00' N the aircraft flies 600 nm west. The final change in longitude can be calculated using the same departure equation:

Change in longitude = 600 nm / Cos 04° 00' which = 601.47' W.

Dividing this by 60 gives change in longitude = 10° 1.47'W, so the longitude at the final longitude at the end of the fourth leg is equal to the longitude at the end of the third leg (19° 56.7' W) + the change of longitude during the fourth leg (10° 1.47'W) which gives final longitude of 29° 58.17' W, which is closest to (option c).

CIRCLES 8 c.

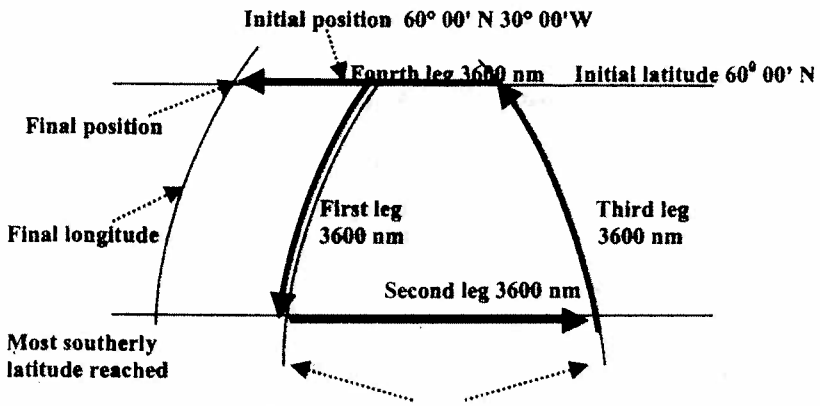
The difference between the initial great circle track and the mean great circle track is the conversion angle. This can be calculated using the equation:

Conversion angle = ½ Change in longitude x Sin of the mean latitude

So for any given change in longitude the conversion angle will be greatest where the sin of the change in latitude is greatest. Of the options given in this question option a gives sin 27.5 = 0.46, option b gives sin 0 = 0, option c gives sin 57.5 = 0.84, and option d gives sin 0 = 0. So option c, give the greatest difference between initial and mean great circle track.

CIRCLES 9 b.

When solving this type of question it is helpful to start with a sketch of the situation as illustrated below.



Meridian converge as to the North and South of the equator

To calculate the change in latitude during the first leg (Northerly) and third leg (Southerly) use the fact that each nm along any meridian is equal to 1' latitude. So 3600 nm = 3600' which is 60° change of latitude.

So the most Southerly latitude reached is 60° South of the initial latitude of (60° 00' N). The most Southerly latitude is therefore the Equator.

During the second leg the aircraft flies 3600 nm to the East at latitude 00° 00' S. Using the departure equation to calculate the change in longitude gives:

Departure = Change in longitude x Cos latitude

Which can be rearranged to give: Change in longitude = Departure/Cos latitude

Inserting the data above gives:

Change in longitude = 3600 nm / Cos 00° 00' which = 3600' E.

Dividing this by 60 gives change in longitude = 60° E so the longitude at the end of the second leg:

Longitude = initial longitude (30° 00'W) + (60° 00' E) = 30° 00' E

During the third the aircraft flies 3600 nm North taking it back to latitude 60° N but now at longitude 30° 00' E.

During the fourth leg at latitude 60° 00' N the aircraft flies 3600 nm west. The final change in longitude can be calculated using the same departure equation:

Change in longitude = 3600 nm / Cos 60° 00' which = 7200' W.

Dividing this by 60 gives change in longitude = 120° 00'W, so the longitude at the final longitude at the end of the fourth leg is equal to the longitude at the end of the third leg (30° 00' E) + the change of longitude during the fourth leg (120° 00' W) which gives a final longitude of 90° 00' W.

So the final position is 60° 00' N 90° 00' W (option b).

CIRCLES 10 d.

The great circle bearing from any point to any other point is equal to the Rhumb line bearing plus the conversion angle. This can be calculated using the equation:

Conversion angle = ½ Change in longitude x Sin of the mean latitude

The change in longitude in this question is from 060° E to 030° W, which is a change of 90°

Both points are at 70° S, so the change in latitude is nil, so the mean latitude is 70° S.

Inserting these figures into the conversion angle equation gives:

Conversion angle = ½ x 90° x Sin 70° which = 42.3°

The rhumb line bearing from point A to point B is 090°, so the great circle bearing is 090° + 42.3° = 132.3° or approximately 135° (option d).

CIRCLES 11 c.

The distance between any two meridians on any specified parallel of latitude is called the departure. This can be calculated using the standard equation:

Departure (in nm) = Change in longitude x Cos latitude

This can be rearranged to give:

Change in longitude = Departure (in nm) / Cos latitude

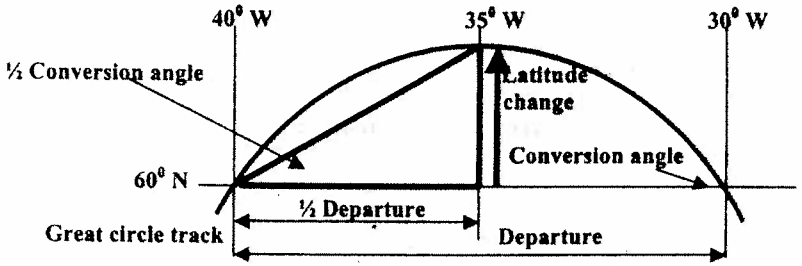
To use this equation the distance given in the question must be converted from km into nm. This can be done using the CRP5 or by using the conversion factor 1.85 km = 1 nm to give 315 km / 1.85 = 170.27 nm

The latitude in this question is constant at 60° N so putting these figures into the above equation gives:

Change in longitude = 170.27 nm / Cos 60 = 340.54' which is 5.68°.

The aircraft is flying East so the new longitude will be the initial longitude – the change in longitude.

So new longitude = 005° W-5.68°. = 0.68° E which is approximately 000° 40' E (option c).

CIRCLES 12 a.

To solve this type of problem it is helpful to make a sketch of the situation similar to the one below.